Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 20

Bijections, Inverses, and Composition

Covers Chartrand & Zhang, Section 5.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Bijective means invertible

The equivalence to keep in mind

$f : A \to B$ has an inverse **if and only if** f is bijective.
Onto guarantees $f^{-1}(b)$ is defined for every b. **One-to-one** guarantees it is defined *uniquely*. Those are exactly the two conditions for f^{-1} to be a function.

Finding an inverse

Write $y = f(x)$, solve for x in terms of y, and that expression is $f^{-1}(y)$. If you cannot solve uniquely, the function was not bijective.

Three mistakes to expect

1. **Writing f^{-1} for a non-bijective function.** It does not exist.
2. **Confusing f^{-1} with $1/f$.** Different things entirely.
3. **Assuming $g \circ f = f \circ g$.** Composition is not commutative.

Part II: Practice Problems

Section A — Bijections and inverses

1. For each function, decide whether it is bijective. If it is, give f^{-1}; if not, say which property fails and give a witness.

(a) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 6 - 2x$.

(b) $g : \mathbb{Z} \to \mathbb{Z}$, $g(n) = n^2$.

(c) $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^3 + 1$.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 4x - 9$. Verify directly that $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Section B — Composition

3. Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x - 1$ and $g(x) = x + 3$.

(a) Find $g \circ f$ and $f \circ g$.

(b) Is $g \circ f = f \circ g$? If not, find where they agree.

(c) Find $(g \circ f)^{-1}$ and compare it with $f^{-1} \circ g^{-1}$.

4. Let $f : A \to B$ and $g : B \to C$ both be one-to-one. Prove that $g \circ f$ is one-to-one.

Section C — The induced equivalence relation

5. Let $f : \{1, 2, 3, 4, 5, 6\} \to \{a, b, c\}$ be $f = \{(1, a), (2, b), (3, a), (4, c), (5, b), (6, a)\}$. Define $x \sim y$ if $f(x) = f(y)$.

(a) List the equivalence classes of $\sim$.

(b) Is f one-to-one? How do the classes show it?

(c) Is f onto? How do the classes show it?

Section D — Writing

6. In five to eight complete sentences, explain why a function must be both one-to-one and onto to have an inverse. Address each property separately, and connect your answer to what it takes for a relation to be a function at all.

Section E — Challenge (optional)

7. Prove that if $f : A \to B$ and $g : B \to C$ are both bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.